John Pelan

johnpelan1@gmail.com | https:/www.linkedin.com/in/john-pelan12/ | https://jpeelan.github.io/

EDUCATION

Purdue University West Lafayette, IN
Bachelor of Science in Electrical Engineering Expected: May 2028
Concentrations in Automatic Controls, Artificial Intelligence and Machine Learning
Minor in Mathematics

WORK EXPERIENCE

Swinergy Minneapolis, MN
Electrical Engineering Intern June 2026-August 2026

- Designed and prototyped a wireless ESP32-S3 sensor and actuator network for remote monitoring and control of a pilot-scale multi-stage digestion reactor
- Integrated industrial sensors, including 4–20 mA flow and level sensors, Type-K thermocouples, pH electrodes, and analog measurement circuits
- Developed a 24 V power and control architecture incorporating DC-DC conversion, circuit protection, MOSFET-driven relays, and pump and solenoid control

Purdue University College of Engineering West Lafayette, IN
Undergraduate Teaching Assistant January 2026-May 2026

- Guide students through hands-on electrical engineering labs involving circuit design, breadboard implementation, and troubleshooting using oscilloscopes, function generators, and power supplies
- Diagnose and resolve circuit faults by validating measurements, identifying wiring and design errors, and explaining underlying circuit behavior
- Provide clear, constructive technical feedback on lab work while reinforcing proper lab practices and measurement techniques
- Maintain professional communication with students and course staff while adapting instruction in real time

TECHNICAL EXPERIENCE

Purdue's Mechanisms and Robotic Systems ("MARS") Lab West Lafayette, IN
Undergraduate Researcher January 2026-Present

- Develop and prototype tactile sensing systems for robotic applications across multiple sensing modalities, including piezoresistive, capacitive, and photodiode-based architectures, with an emphasis on multi-point force and contact detection.
- Design custom rigid and flexible PCBs, sensor interfaces, multiplexed signal-routing architectures, and embedded data-acquisition hardware to support compact, high-density tactile sensor arrays.
- Develop flexible fingertip sensors and real-time tactile visualization systems for robotic platforms, including integration with the LEAP Hand and LED-based spatial feedback for observing sensor response during physical interaction.

Purdue Humanoid Robotics Club West Lafayette, IN
Member August 2025-Present

- Collaborate with a multidisciplinary team to design and build a humanoid robot, focusing on developing precise electrical schematics for the robot's arms and legs to enable advanced articulation and movement

SKILLS

- Programming: Proficient in Python, C, and Matlab
- Circuit Design Software: Proficient in KiCad, Altium, and LTspice
- Hardware Description Languages: Proficient in System Verilog for digital design
- Electronics Testing & Measurement: Proficient with oscilloscopes, digital multimeters (DMMs), waveform/function generators, and bench power supplies
- Prototyping and Research: Proficient in soldering and prototyping circuits